

Empirical Study of Memorization in Differentially Privately Fine-Tuned Large Language Models

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17 Apr 2026

[Link to GitHub Repo](#)

Abstract

1. Introduction

2. Background

Differential privacy (DP) is a mathematical framework that leverages probability theories to rigorously quantify privacy in algorithms (Dwork et al., 2006). Formally, let $\mathcal{X}$ denote the data space and $X = (x_1, \dots, x_n) \in \mathcal{X}^n$ be a dataset. We can then define another dataset $X' = (x'_1, \dots, x'_n) \in \mathcal{X}^n$ to be a *neighboring dataset* of X , denoted as $X \sim X'$, if there exists $i \in \{1, \dots, n\}$ such that $x_i \neq x'_i$, and for all $j \neq i$, $x_j = x'_j$. With this setup, we say that a randomized algorithm $\mathcal{M}$ is (ϵ, δ) -*differentially private* if for all datasets $X \sim X'$, and all $S \subseteq \Omega$ where Ω is the output space of $\mathcal{M}$,

$$\mathbb{P}(\mathcal{M}(X) \in S) \leq e^\epsilon \mathbb{P}(\mathcal{M}(X') \in S) + \delta.$$

In the context of training deep neural networks, we can adapt this definition and make gradient descent a differentially private algorithm so that the trained models do not memorize specific details of any training data. Defining $\mathcal{L}$ as the loss function, α as the learning rate, instead of updating the weights by $\alpha \nabla \mathcal{L}$ at every step, we first compute each per-sample gradient $\nabla \mathcal{L}_i$ and clip it to the L2 norm bound C , then sum the clipped per-sample gradients and add random noise from $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ to the sum before updating. By choosing the parameters σ and C strategically, one can show that this algorithm, named *differentially private stochastic gradient descent* (DP-SGD), satisfies (ϵ, δ) -differential privacy (Abadi et al., 2016).

3. Related Work

4. Methodology

5. Results & Discussion

6. Conclusion

6.1. Limitations & Future Directions

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